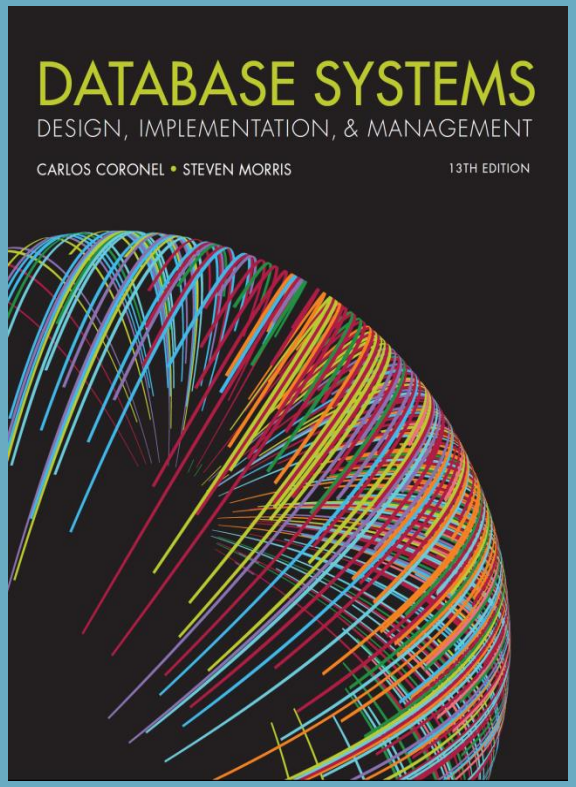


Chapter 7

Introduction to Structured Query Language (SQL)



Learning Objectives

- After completing this chapter, you will be able to:
  - Retrieve specified columns of data from a database
  - Join multiple tables in a single SQL query
  - Restrict data retrievals to rows that match complex criteria
  - Aggregate data across groups of rows
  - Create subqueries to preprocess data for inclusion in other queries
  - Identify and use a variety of SQL functions for string, numeric, and date manipulation
  - Explain the key principles in crafting a SELECT query

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Introduction to SQL (1 of 4)

- Categories of SQL functions
  - Data definition language (DDL)
  - Data manipulation language (DML)
  - Transaction control language (TCL)
  - Data control language (DCL)
- SQL is relatively easy to learn
  - Nonprocedural language with basic command vocabulary set of less than 100 words
  - Differences in SQL dialects are minor

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Introduction to SQL (2 of 4)

Table 7.2	SQL Data Definition Commands	
Command or Option	Description	Covered
CREATE SCHEMA AUTHORIZATION	**Creates a database schema**	**Chapter 8**
CREATE TABLE	**Creates a new table in the user's database schema**	**Chapter 8**
NOT NULL	Ensures that a column will not have null values	Chapter 8
UNIQUE	Ensures that a column will not have duplicate values	Chapter 8
PRIMARY KEY	Defines a primary key for a table	Chapter 8
FOREIGN KEY	Defines a foreign key for a table	Chapter 8
DEFAULT	Defines a default value for a column (when no value is given)	Chapter 8
CHECK	Validates data in an attribute	Chapter 8
CREATE INDEX	**Creates an index for a table**	**Chapter 8**
CREATE VIEW	**Creates a dynamic subset of rows and columns from one or more tables**	**Chapter 8**
ALTER TABLE	**Modifies a table's definition (adds, modifies, or deletes attributes or constraints)**	**Chapter 8**
CREATE TABLE AS	**Creates a new table based on a query in the user's database schema**	**Chapter 8**
DROP TABLE	**Permanently deletes a table (and its data)**	**Chapter 8**
DROP INDEX	**Permanently deletes an index**	**Chapter 8**
DROP VIEW	**Permanently deletes a view**	**Chapter 8**

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Introduction to SQL (3 of 4)

Table 7.3	Other SQL Commands	
Command or Option	**Description**	**Covered**
Transaction Control Language		
COMMIT	Permanently saves data changes	Chapter 8
ROLLBACK	Restores data to its original values	Chapter 8
Data Control Language		
GRANT	Gives a user permission to take a system action or access a data object	Chapter 16
REVOKE	Removes a previously granted permission from a user	Chapter 16

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Introduction to SQL (4 of 4)

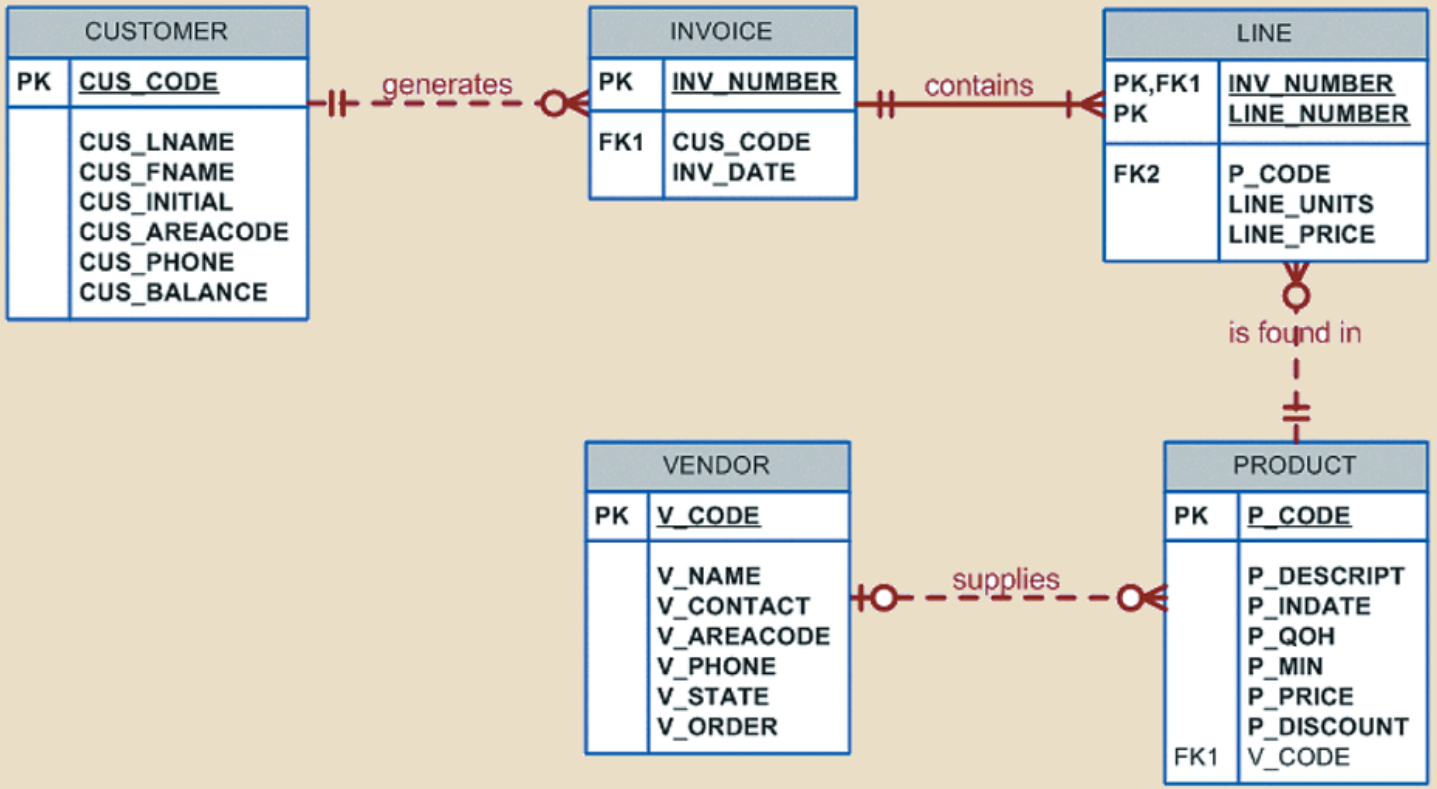
- Data type: specification about the kinds of data that can be stored in an attribute
  - Influence queries that retrieve data
- Fundamental types of data
  - Character data
  - Numeric data
  - Date data
- At the heart of SQL is the query
  - Covers both questions and actions

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The Database Model

FIGURE 7.1 THE DATABASE MODEL



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Business Rules

1. A customer may generate many invoices. Each invoice is generated by one customer.
2. An invoice contains one or more invoice lines. Each invoice line is associated with one invoice.
3. Each invoice line references one product. A product may be found in many invoice lines
4. A vendor may supply many products. Some vendors do not supply products (yet to be approved)
5. If a product is vendor-supplied, that product is supplied by only a single vendor
6. Some products are produced in-house, i.e. they are not supplied by a vendor

DDL --- VENDOR and PRODUCT

VENDOR

```
CREATE TABLE VENDOR (
V_CODE                  INTEGER,
V_NAME                  VARCHAR(35) NOT NULL,
V_CONTACT               VARCHAR(15) NOT NULL,
V_AREACODE              CHAR(3) NOT NULL,
V_PHONE                 CHAR(8) NOT NULL,
V_STATE                 CHAR(2) NOT NULL,
V_ORDER                 CHAR(1) NOT NULL,
PRIMARY KEY (V_CODE));
```

PRODUCT

```
CREATE TABLE PRODUCT (
P_CODE                  VARCHAR2(10) CONSTRAINT PRODUCT_P_CODE_PK PRIMARY KEY,
P_DESCRIPT              VARCHAR2(35) NOT NULL,
P_INDATE  DATE NOT NULL,
P_QOH                   NUMBER NOT NULL,
P_MIN                   NUMBER NOT NULL,
P_PRICE    NUMBER(8,2) NOT NULL,
P_DISCOUNT              NUMBER(5,2) NOT NULL,
V_CODE                  NUMBER,
CONSTRAINT PRODUCT_V_CODE_FK FOREIGN KEY (V_CODE) REFERENCES VENDOR);
```

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DDL --- CUSTOMER

CUSTOMER

```
CREATE TABLE CUSTOMER (
CUS_CODE            NUMBER PRIMARY KEY,
CUS_LNAME           VARCHAR(15) NOT NULL,
CUS_FNAME           VARCHAR(15) NOT NULL,
CUS_INITIAL         CHAR(1),
CUS_AREACODE        CHAR(3) DEFAULT '615' NOT NULL CHECK(CUS_AREACODE IN ('615','713','931')),
CUS_PHONE           CHAR(8) NOT NULL,
CUS_BALANCE         NUMBER(9,2) DEFAULT 0.00,
CONSTRAINT CUS_UI1 UNIQUE(CUS_LNAME,CUS_FNAME));
```

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DDL --- INVOICE and LINE

INVOICE

```
INV_NUMBER          NUMBER PRIMARY KEY,
CUS_CODE            NUMBER    NOT NULL REFERENCES CUSTOMER(CUS_CODE),
INV_DATE            DATE DEFAULT SYSDATE NOT NULL,
CONSTRAINT INV_CK1 CHECK (INV_DATE > TO_DATE('01-JAN-2012','DD-MON-YYYY')));
```

LINE

```
CREATE TABLE LINE (
INV_NUMBER          NUMBER NOT NULL,
LINE_NUMBER         NUMBER(2,0) NOT NULL,
P_CODE              VARCHAR(10) NOT NULL,
LINE_UNITS          NUMBER(9,2) DEFAULT 0.00 NOT NULL,
LINE_PRICE          NUMBER(9,2) DEFAULT 0.00 NOT NULL,
PRIMARY KEY (INV_NUMBER,LINE_NUMBER),
FOREIGN KEY (INV_NUMBER) REFERENCES INVOICE ON DELETE CASCADE,
FOREIGN KEY (P_CODE) REFERENCES PRODUCT(P_CODE),
CONSTRAINT LINE_UI1 UNIQUE(INV_NUMBER, P_CODE));
```


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DDL --- Constraints

VENDOR

```
CREATE TABLE VENDOR (
V_CODE          INTEGER,
V_NAME          VARCHAR(35),
V_CONTACT       VARCHAR(15),
V_AREACODE      CHAR(3),
V_PHONE         CHAR(8),
V_STATE         CHAR(2),
V_ORDER         CHAR(1)
);
```

is337002Conn | VENDOR

Columns | Data | Constraints | Grants | Sta

Actions...

Column Name	Data Type
V_CODE	NUMBER
V_NAME	VARCHAR2(35 BYTE)
V_CONTACT	VARCHAR2(15 BYTE)
V_AREACODE	CHAR(3 BYTE)
V_PHONE	CHAR(8 BYTE)
V_STATE	CHAR(2 BYTE)
V_ORDER	CHAR(1 BYTE)

Now add constraints….

```
ALTER TABLE VENDOR
ADD CONSTRAINT PK_VCODE_CONSTRAINT  PRIMARY KEY(V_CODE);

ALTER TABLE VENDOR
MODIFY V_NAME NOT NULL ;
```

Note the difference between ADD and MODIFY

Note the constraint name

```
ALTER TABLE VENDOR
RENAME CONSTRAINT SYS_C00236678 to V_NAME_NN
```

SupermanCon | VENDOR

Columns | Data | Constraints | Grants | Statistics | Triggers | Flashback

Actions...

Constraint Name	Constraint Type	Search Condition
PK_VCODE_CONST...	Primary_Key	(null)
V_NAME_NN	Check	"V_NAME" IS NOT NULL

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DDL --- Constraints

VENDOR

```
CREATE TABLE VENDOR (
V_CODE                  INTEGER,
V_NAME                  VARCHAR(35) NOT NULL,
V_CONTACT               VARCHAR(15) NOT NULL,
V_AREACODE              CHAR(3) NOT NULL,
V_PHONE                 CHAR(8) NOT NULL,
V_STATE                 CHAR(2) NOT NULL,
V_ORDER                 CHAR(1) NOT NULL,
PRIMARY KEY (V_CODE));
```

VENDOR

```
CREATE TABLE VENDOR
(
 V_CODE                  INTEGER     CONSTRAINT PK_VCODE PRIMARY KEY,
 V_NAME                 VARCHAR(35) CONSTRAINT NN_VNAME NOT NULL,
 V_CONTACT               VARCHAR(15) CONSTRAINT NN_VCONTACT NOT NULL,
 V_AREACODE             CHAR(3)     CONSTRAINT NN_VAREACODE NOT NULL,
 V_PHONE                CHAR(8)    CONSTRAINT NN_VPHONE NOT NULL,
 V_STATE                CHAR(2)    CONSTRAINT NN_VSTATE NOT NULL,
 V_ORDER                CHAR(1)     CONSTRAINT NN_VORDER NOT NULL
);
```


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Composite Key

```
CREATE TABLE LINE (
INV_NUMBER      NUMBER NOT NULL,
LINE_NUMBER     NUMBER(2,0) NOT NULL,
P_CODE          VARCHAR(10) NOT NULL,
LINE_UNITS      NUMBER(9,2) DEFAULT 0.00 NOT NULL,
LINE_PRICE      NUMBER(9,2) DEFAULT 0.00 NOT NULL,
PRIMARY KEY (INV_NUMBER,LINE_NUMBER),
FOREIGN KEY (INV_NUMBER) REFERENCES INVOICE ON DELETE CASCADE,
FOREIGN KEY (P_CODE) REFERENCES PRODUCT(P_CODE),
CONSTRAINT LINE_UI1 UNIQUE(INV_NUMBER, P_CODE));
```

Questions??

Q. 1) s constraint definition (a) and (b) same

(a) PRIMARY KEY (INV_NUMBER,LINE_NUMBER)

(b) PRIMARY KEY (LINE_NUMBER, INV_NUMBER)

Q. 2) What is meant by ON DELETE CASCADE

FOREIGN KEY (INV_NUMBER) REFERENCES INVOICE ON DELETE CASCADE,

Questions??

Ans 1). The order of the primary key is important because the indexing starts with the first mentioned attribute, then proceeds to the next attribute

So PRIMARY KEY (INV_NUMBER,LINE_NUMBER)

Will have (1001, 1) (1001, 2) (1002, 1) (1003, 1) (1003, 2) so on…

Ans 2). ON DELETE CASCADE

FOREIGN KEY (INV_NUMBER) REFERENCES INVOICE ON DELETE CASCADE,

Because of the cascade delete, when a record in the INVOICE table is deleted, all records in the LINE table will also be deleted that have the same INV_NUMBER value.



Data Manipulation Commands

INSERT: Command to insert data into table

- Syntax - INSERT INTO tablename VALUES();
- Used to add table rows with NULL and NOT NULL attributes

COMMIT: Command to save changes

- Syntax - COMMIT [WORK];
- Ensures database update integrity

Adding Table Rows

- INSERT
  - Used to enter data into table
  - Syntax: INSERT INTO tablename
    VALUES (value1, value2, … , valueN);

```
INSERT INTO VENDOR VALUES(21225,'Bryson, Inc.'    ,'Smithson','615','223-
  3234','TN','Y');

INSERT INTO VENDOR VALUES(21226,'SuperLoo, Inc.'  ,'Flushing','904','215-
  8995','FL','N');

INSERT INTO VENDOR VALUES(11111,'ABCDEFGHIJK' ,
  'LMNOPQRST','222','222-2222','AA','Y');

COMMIT;
```

is337002Conn | VENDOR

Columns | Data | Constraints | Grants | Statistics | Triggers | Flashback | Dependencies | Details | Indexes | SQL

Sort... | Filter: | Actions...

	V_CODE	V_NAME	V_CONTACT	V_AREACODE	V_PHONE	V_STATE	V_ORDER
1	21225	Bryson, Inc.	Smithson	615	223-3234	TN	Y
2	21226	SuperLoo, ...	Flushing	904	215-8995	FL	N
3	11111	ABCDEFG...	LMNOPQRST	222	222-2222	AA	Y

Adding Dates in Table Rows (cont'd.)

```
INVOICE
INV_NUMBER          NUMBER PRIMARY KEY,
CUS_CODE NUMBER     NOT NULL REFERENCES CUSTOMER(CUS_CODE),
INV_DATE            DATE DEFAULT SYSDATE NOT NULL,
CONSTRAINT INV_CK1 CHECK (INV_DATE > TO_DATE('01-JAN-2012','DD-MON-YYYY')));
```

```
ALTER SESSION SET NLS_DATE_FORMAT = 'MM/DD/YYYY';

INSERT INTO INVOICE VALUES(1001,10014,'12/24/2012');
```

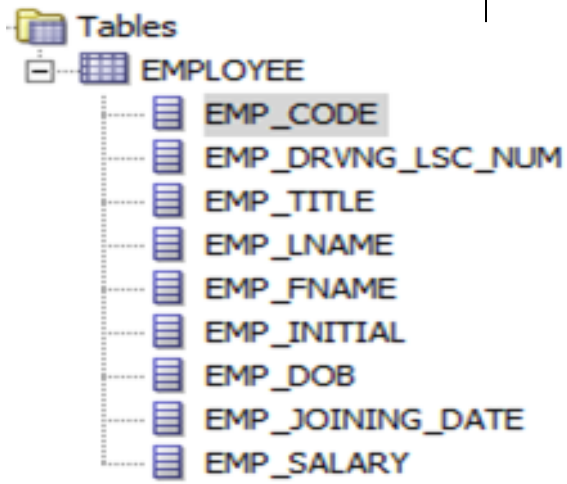
```
Alter SESSION SET NLS_DATE_FORMAT = 'DD-MON-YYYY';

INSERT INTO INVOICE VALUES(1001,10014,'16-JAN-2012');
```

Adding Table Rows (cont'd.)

When adding dates.... e.g., EMP_DOB and EMP_HIRE_DATE

```
ALTER SESSION SET NLS_DATE_FORMAT =
    'MM/DD/YYYY';
INSERT INTO EMPLOYEE VALUES
 (
  1,
  'AAAA',
  'DR.',
  'SMITH',
  'JOHN',
  NULL,
  '10/31/1985',
  '12/24/2002',
  40000
 );
```




```
Alter session set NLS_DATE_FORMAT
    = 'DD-MON-YYYY';
INSERT INTO EMPLOYEE VALUES
 (
  2,
  'BBBB',
  'MS.',
  'OBAMA',
  'MICHELLE',
  NULL,
  '31-OCTOBER -1965',
  '28 - FEBRUARY -2002',
  40000
 );
```

is337002Conn | EMPLOYEE

Columns | Data | Constraints | Grants | Statistics | Triggers | Flashback | Dependencies | Details | Indexes | SQL

Sort... | Filter: | Actions...

	EMP_CODE	EMP_DRVNG_LSC_NUM	EMP_TITLE	EMP_LNAME	EMP_FNAME	EMP_INITIAL	EMP_DOB	EMP_JOIN
1	1	AAAA	DR.	SMITH	JOHN	(null)	31-OCT-1985	24-DEC-2002
2	2	BBBB	MS.	OBAMA	MICHELLE	(null)	31-OCT-1985	28-FEB-2002

Basic SELECT Queries

- Each clause in a SELECT query performs a specific function
  - SELECT: specifies the attributes to be returned by the query
  - FROM: specifies the table(s) from which the data will be retrieved
  - WHERE: filters the rows of data based on provided criteria
  - GROUP BY: groups the rows of data into collections based on sharing the same values in one or more attributes
  - HAVING: filters the groups formed in the GROUP BY clause based on provided criteria
  - ORDER BY: sorts the final query result rows in ascending or descending order based on the values of one or more attributes
- SQL commands can be grouped together on a single line
  - Complex command sequences are best shown on separate lines, with space between the SQL command and the command's components

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SELECT Statement Options (1 of 8)

- The SELECT query specifies the columns to be retrieved as a column list
  - Syntax:

    SELECT *columnlist*

    FROM *tablelist;*
  - The columnlist represents one or more attributes, separated by commas
  - A wildcard character is a symbol that can be used as a general substitute for other characters or commands
- Using column aliases
  - Alternative name for a column or table in a SQL statement
- Using computed columns
  - Computed column (also called a calculated column) represents a derived attribute
- Arithmetic operators: the rule of precedence
  - Rules that establish the order in which computations are completed

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SELECT Statement Options (2 of 8)

SELECT * FROM PRODUCT;

FIGURE 7.2 SELECT AN ENTIRE TABLE

P_CODE	P_DESCRIPT	P_INDATE	P_QOH	P_MIN	P_PRICE	P_DISCOUNT	V_CODE
11QER/31	Power painter, 15 psi., 3-nozzle	03-Nov-17	8	5	109.99	0.00	25595
13-Q2/P2	7.25-in. pwr. saw blade	13-Dec-17	32	15	14.99	0.05	21344
14-Q1/L3	9.00-in. pwr. saw blade	13-Nov-17	18	12	17.49	0.00	21344
1546-QQ2	Hrd. cloth, 1/4-in., 2x50	15-Jan-18	15	8	39.95	0.00	23119
1558-QVV1	Hrd. cloth, 1/2-in., 3x50	15-Jan-18	23	5	43.99	0.00	23119
2232/QTY	B&D jigsaw, 12-in. blade	30-Dec-17	8	5	109.92	0.05	24288
2232/QWE	B&D jigsaw, 8-in. blade	24-Dec-17	6	5	99.87	0.05	24288
2238/QPD	B&D cordless drill, 1/2-in.	20-Jan-18	12	5	38.95	0.05	25595
23109-HB	Claw hammer	20-Jan-18	23	10	9.95	0.10	21225
23114-AA	Sledge hammer, 12 lb.	02-Jan-18	8	5	14.40	0.05	
54778-2T	Rat-tail file, 1/8-in. fine	15-Dec-17	43	20	4.99	0.00	21344
89-WRE-Q	Hicut chain saw, 16 in.	07-Feb-18	11	5	256.99	0.05	24288
PVC23DRT	PVC pipe, 3.5-in., 8-ft	20-Feb-18	188	75	5.87	0.00	
SM-18277	1.25-in. metal screw, 25	01-Mar-18	172	75	6.99	0.00	21225
SW-23116	2.5-in. wd. screw, 50	24-Feb-18	237	100	8.45	0.00	21231
WR3/TT3	Steel matting, 4'x8'x1/6", .5" mesh	17-Jan-18	18	5	119.95	0.10	25595

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SELECT Statement Options (3 of 8)

```
SELECT P_CODE, P_DESCRIPT, P_PRICE, P_QOH FROM PRODUCT;
```

FIGURE 7.3 SELECT WITH A COLUMN LIST

P_CODE	P_DESCRIPT	P_PRICE	P_QOH
11QER/31	Power painter, 15 psi., 3-nozzle	109.99	8
13-Q2/P2	7.25-in. pwr. saw blade	14.99	32
14-Q1/L3	9.00-in. pwr. saw blade	17.49	18
1546-QQ2	Hrd. cloth, 1/4-in., 2x50	39.95	15
1558-QVV1	Hrd. cloth, 1/2-in., 3x50	43.99	23
2232/QTY	B&D jigsaw, 12-in. blade	109.92	8
2232/QWE	B&D jigsaw, 8-in. blade	99.87	6
2238/QPD	B&D cordless drill, 1/2-in.	38.95	12
23109-HB	Claw hammer	9.95	23
23114-AA	Sledge hammer, 12 lb.	14.40	8
54778-2T	Rat-tail file, 1/8-in. fine	4.99	43
89-WRE-Q	Hicut chain saw, 16 in.	256.99	11
PVC23DRT	PVC pipe, 3.5-in., 8-ft	5.87	188
SM-18277	1.25-in. metal screw, 25	6.99	172
SW-23116	2.5-in. wd. screw, 50	8.45	237
WR3/TT3	Steel matting, 4'x8'x1/6", .5" mesh	119.95	18

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SELECT Statement Options (4 of 8)

SELECT p_code, p_descript AS description, p_price AS "unit price", p_qoh qty FROM product;

FIGURE 7.4 SELECT WITH COLUMN ALIASES

P_CODE	DESCRIPTION	Unit Price	QTY
11QER/31	Power painter, 15 psi., 3-nozzle	109.99	8
13-Q2/P2	7.25-in. pwr. saw blade	14.99	32
14-Q1/L3	9.00-in. pwr. saw blade	17.49	18
1546-QQ2	Hrd. cloth, 1/4-in., 2x50	39.95	15
1558-QVV1	Hrd. cloth, 1/2-in., 3x50	43.99	23
2232/QTY	B&D jigsaw, 12-in. blade	109.92	8
2232/QWE	B&D jigsaw, 8-in. blade	99.87	6
2238/QPD	B&D cordless drill, 1/2-in.	38.95	12
23109-HB	Claw hammer	9.95	23
23114-AA	Sledge hammer, 12 lb.	14.40	8
54778-2T	Rat-tail file, 1/8-in. fine	4.99	43
89-WRE-Q	Hicut chain saw, 16 in.	256.99	11
PVC23DRT	PVC pipe, 3.5-in., 8-ft	5.87	188
SM-18277	1.25-in. metal screw, 25	6.99	172
SW-23116	2.5-in. wd. screw, 50	8.45	237
WR3/TT3	Steel matting, 4'x8'x1/6", .5" mesh	119.95	18

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SELECT Statement Options (5 of 8)

Table 7.4: The Arithmetic Operators	
Operator	**Description**
+	Add
-	Subtract
*	Multiply
/	Divide
^	Raise to the power of (some applications use ** instead of ^)

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COMPUTED COLUMNS and ALIASES(6 of 8)

- -- Using computed columns and aliases
- SELECT P_DESCRIPT, P_QOH, P_PRICE, P_QOH*P_PRICE FROM PRODUCT;
- SELECT P_DESCRIPT, P_QOH, P_PRICE, P_QOH*P_PRICE AS TOTVALUE FROM PRODUCT;
- SELECT P_CODE, P_INDATE, P_INDATE + 90 as EXPDATE FROM PRODUCT;
- SELECT P_CODE, P_INDATE, SYSDATE - 90 AS CUTOFF FROM PRODUCT;

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SELECT Statement Options (7 of 8)

- Date arithmetic
  - Values are stored as a number of days; it is possible to perform date arithmetic in a query
- Listing unique values
  - SQL's DISTINCT clause produces a list of only those values that are different from one another
  - Command example:

```
SELECT    DISTINCT V_CODE
FROM      PRODUCT;
```

FIGURE 7.7 A LISTING OF DISTINCT V_CODE VALUES IN THE PRODUCT TABLE

V_CODE
21225
21231
21344
23119
24288
25595

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SELECT Statement Options (8 of 8)

```
SELECT V_NAME "Vendor Name",
            V_CONTACT "Vendor Contact",
            V_AREACODE "Vendor Area Code",
             V_PHONE "Phone Number"
FROM VENDOR
WHERE V_AREACODE = '615';
```

```
Vendor Name                       Vendor Contact  Ven Phone Nu
--------------------------------- --------------- --- --------
Bryson, Inc.                      Smithson        615 223-3234
DeeEee Supply                     Singh           615 228-3245
Gomez Bros.                       Ortega          615 889-2546
Brackman Bros.                    Browning        615 228-1410
ORDVA, Inc.                       Hakford         615 898-1234
Damal Supplies                    Smythe          615 890-3529

6 rows selected.
```

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FROM Clause Options

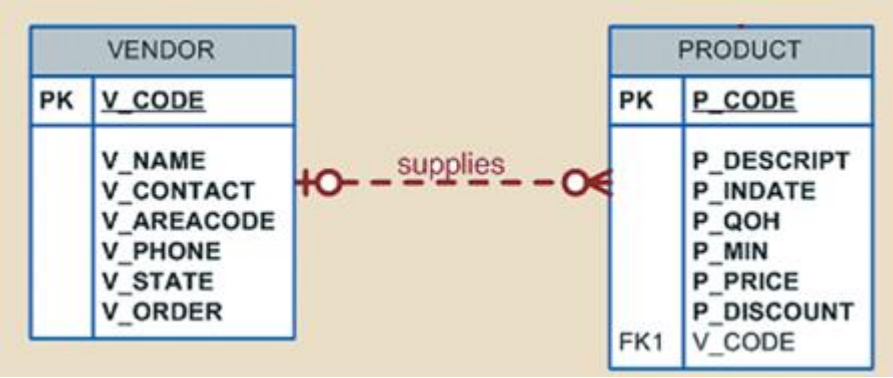
- FROM clause of the query specifies the table or tables from which the data is to be retrieved
  - Inner joins return only rows from the tables that match on a common value
  - Outer joins return the same matched rows as the inner join, plus unmatched rows from one table or the other
- Natural join returns all rows with matching values in the matching columns and eliminates duplicate columns
  - Determines the common attribute(s) by looking for attributes with identical names and compatible data types
  - Selects only the rows with common values in the common attribute(s)
  - If there are no common attributes, returns the relational product of the two tables
  - Syntax:

    SELECT *column-list* FROM *table1* NATURAL JOIN *table2*

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FROM Clause Options

Table 7.5 Creating Links through Foreign Keys		
Table	**Attributes To Be Shown**	**Linking Attribute**
PRODUCT	P_DESCRIPT, P_PRICE	V_CODE
VENDOR	V_NAME, V_CONTACT, V_AREACODE, V_PHONE	V_CODE



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FROM Clause Options

- JOIN USING syntax
  - Returns only the rows with matching values in the column indicated in the USING clause—and that column must exist in both tables
  - Syntax:

    SELECT *column-list* FROM *table1* JOIN *table2* USING (*common-column*)
- JOIN ON syntax
  - Express a join when the tables have no common attribute names
  - Query returns only the rows that meet the indicated join condition
  - Syntax:

    SELECT *column-list* FROM *table1* JOIN table2 ON *join-condition*
- Common attribute names
  - Most common cause of duplicate column names is the existence of a foreign key

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Result of a Join between two tables

```
SELECT P_DESCRIPT, P_PRICE, V_NAME, V_CONTACT, V_AREACODE, V_PHONE
FROM PRODUCT, VENDOR
WHERE PRODUCT.V_CODE = VENDOR.V_CODE;
```

```
SELECT P_DESCRIPT, P_PRICE, V_NAME, V_CONTACT, V_AREACODE, V_PHONE
FROM PRODUCT JOIN VENDOR ON (PRODUCT.V_CODE = VENDOR.V_CODE);
```

```
SELECT P_DESCRIPT, P_PRICE, V_NAME, V_CONTACT, V_AREACODE, V_PHONE
FROM PRODUCT JOIN VENDOR
using (V_CODE);
```

```
SELECT P_DESCRIPT, P_PRICE, V_NAME, V_CONTACT, V_AREACODE, V_PHONE
FROM PRODUCT NATURAL JOIN VENDOR;
```

	P_DESCRIPT	P_PRICE	V_NAME	V_CONTACT	V_AREACODE	V_PHONE
1	Claw hammer	9.95	Bryson, Inc.	Smithson	615	223-3234
2	1.25-in. metal screw, 25	6.99	Bryson, Inc.	Smithson	615	223-3234
3	2.5-in. wd. screw, 50	8.45	Drills Supply	Singh	615	228-3245
4	7.25-in. pwr. saw blade	14.99	Gomez Bros.	Ortega	615	889-2546
5	9.00-in. pwr. saw blade	17.49	Gomez Bros.	Ortega	615	889-2546
6	Rat-tail file, 1/8-in. fine	4.99	Gomez Bros.	Ortega	615	889-2546
7	Hrd. cloth, 1/4-in., 2x50	39.95	Randsets Ltd.	Anderson	901	678-3998

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Result of a Join between two tables

```
SELECT PRODUCT.P_DESCRIPT, PRODUCT.P_PRICE, VENDOR.V_NAME, VENDOR.V_CONTACT,
VENDOR.V_AREACODE, VENDOR.V_PHONE
FROM PRODUCT, VENDOR
WHERE PRODUCT.V_CODE = VENDOR.V_CODE
ORDER BY PRODUCT.P_PRICE;
```

	P_DESCRIPT	P_PRICE	V_NAME	V_CONTACT	V_AREACODE	V_PHONE
1	Rat-tail file, 1/8-in. fine	4.99	Gomez Bros.	Ortega	615	889-2546
2	1.25-in. metal screw, 25	6.99	Bryson, Inc.	Smithson	615	223-3234
3	2.5-in. wd. screw, 50	8.45	Drills Supply	Singh	615	228-3245
4	Claw hammer	9.95	Bryson, Inc.	Smithson	615	223-3234
5	7.25-in. pwr. saw blade	14.99	Gomez Bros.	Ortega	615	889-2546
6	9.00-in. pwr. saw blade	17.49	Gomez Bros.	Ortega	615	889-2546
7	BeeDee cordless drill, 1/2-in.	38.95	Rubicon Systems	Orton	904	456-0092

```
SELECT P.P_DESCRIPT, P.P_PRICE, V.V_NAME, V.V_CONTACT, V.V_AREACODE, V.V_PHONE
FROM PRODUCT P, VENDOR V
WHERE P.V_CODE = V.V_CODE
ORDER BY P.P_PRICE;
```

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Joining Tables With an Alias

- An alias may be used to identify the source table from which the data is taken
- The ability to specify a table alias is very useful
  - Using a table alias allows the database programmer to improve the maintainability of the code by using a table alias that is descriptive of what data the table is providing within the query

- Add alias after table name in FROM clause
  - FROM tablename alias

Table name: AGENT

AGENT_CODE	AGENT_PHONE
125	6152439887
167	6153426778
231	6152431124
333	9041234445

Table name: CUSTOMER

CUS_CODE	CUS_LNAME	CUS_ZIP	AGENT_CODE
1132445	Walker	32145	231
1217782	Adares	32145	125
1312243	Rakowski	34129	167
1321242	Rodriguez	37134	125
1542311	Smithson	37134	421
1657399	Vanloo	32145	231

```
SELECT C.CUS_CODE, C.CUS_LNAME, C.CUS_ZIP, A.AGENT_CODE, A.AGENT_PHONE
FROM CUSTOMER C JOIN AGENT A
ON C.AGENT_CODE  = A.AGENT_CODE;
```

Join of Three Tables

```
SELECT I.INV_NUMBER, P.P_CODE, P.P_DESCRIPT, L.LINE_UNITS, L.LINE_PRICE
FROM INVOICE I JOIN LINE L ON I.INV_NUMBER= L.INV_NUMBER
JOIN PRODUCT P on L.P_CODE = P.P_CODE;
```

FIGURE 7.12 JOIN ON RESULTS

INV_NUMBER	P_CODE	P_DESCRIPT	LINE_UNITS	LINE_PRICE
1001	13-Q2/P2	7.25-in. pwr. saw blade	1	14.99
1001	23109-HB	Claw hammer	1	9.95
1002	54778-2T	Rat-tail file, 1/8-in. fine	2	4.99
1003	2238/QPD	B&D cordless drill, 1/2-in.	1	38.95
1003	1546-QQ2	Hrd. cloth, 1/4-in., 2x50	1	39.95
1003	13-Q2/P2	7.25-in. pwr. saw blade	5	14.99
1004	54778-2T	Rat-tail file, 1/8-in. fine	3	4.99
1004	23109-HB	Claw hammer	2	9.95
1005	PVC23DRT	PVC pipe, 3.5-in., 8-ft	12	5.87
1006	SM-18277	1.25-in. metal screw, 25	3	6.99
1006	2232/QTY	B&D jigsaw, 12-in. blade	1	109.92
1006	23109-HB	Claw hammer	1	9.95
1006	89-WRE-Q	Hicut chain saw, 16 in.	1	256.99
1007	13-Q2/P2	7.25-in. pwr. saw blade	2	14.99
1007	54778-2T	Rat-tail file, 1/8-in. fine	1	4.99
1008	PVC23DRT	PVC pipe, 3.5-in., 8-ft	5	5.87
1008	WR3/TT3	Steel matting, 4'x8'x1/6", .5" mesh	3	119.95
1008	23109-HB	Claw hammer	1	9.95

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Join of Four Tables

- If you want to list the customer's last name, invoice number, invoice date, and product descriptions for all invoices for customer 10014,
- Tables: Customer, Invoice, Product and LINE

```
SELECT C.CUS_LNAME, I.INV_NUMBER, I.INV_DATE, P.P_DESCRIPT
FROM CUSTOMER C, INVOICE I, LINE L, PRODUCT P
WHERE C.CUS_CODE = I.CUS_CODE
AND I.INV_NUMBER = L.INV_NUMBER
AND L.P_CODE = P.P_CODE
AND C.CUS_CODE = 10014
ORDER BY INV_NUMBER;
```

	CUS_LNAME	INV_NUMBER	INV_DATE	P_DESCRIPT
1	Orlando	1001	16/01/08	7.25-in. pwr. saw blade
2	Orlando	1001	16/01/08	Claw hammer
3	Orlando	1006	17/01/08	BeeDee jigsaw, 12-in. blade
4	Orlando	1006	17/01/08	Claw hammer
5	Orlando	1006	17/01/08	Hicut chain saw, 16 in.
6	Orlando	1006	17/01/08	1.25-in. metal screw, 25

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FROM Clause Options (Refer to Week 3- appendix slides)

- Outer joins
  - Returns not only the rows matching the join condition (rows with matching values in the common columns) and returns the rows with unmatched values
  - ANSI standard defines three types of outer joins: left, right, and full

  - SELECT column-list

  FROM table1 LEFT[OUTER] JOIN table2 ON join-condition

- Cross join
  - Performs a relational product (also known as the *Cartesian product*) of two tables

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Recursive Joins

```
CREATE TABLE EMP (
EMP_NUM      NUMBER  PRIMARY KEY,
EMP_TITLE    CHAR(10),
EMP_LNAME    VARCHAR(15) NOT NULL,
EMP_FNAME    VARCHAR(15) NOT NULL,
EMP_INITIAL  CHAR(1),
EMP_DOB      DATE,
EMP_HIRE_DATE    DATE,
EMP_AREACODE     CHAR(3),
EMP_PHONE    CHAR(8),
EMP_MGR      NUMBER);
```

	EMP_NUM	EMP_TITLE	EMP_LNAME	EMP_FNAME	EMP_INITIAL	EMP_DOB	EMP_HIRE_DATE	EMP_AREACODE	EMP_PHONE	EMP_MGR
1	100	Mr.	Kolmycz	George	D	15/06/42	15/03/85	615	324-5456	(null)
2	101	Ms.	Lewis	Rhonda	G	19/03/65	25/04/86	615	324-4472	100
3	102	Mr.	Vandam	Rhett	(null)	14/11/58	20/12/90	901	675-8993	100
4	103	Ms.	Jones	Anne	M	16/10/74	28/08/94	615	898-3456	100
5	104	Mr.	Lange	John	P	08/11/71	20/10/94	901	504-4430	105
6	105	Mr.	Williams	Robert	D	14/03/75	08/11/98	615	890-3220	(null)

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FROM Clause Options (8 of 8)

Recursive joins

Recursive query: joins a table to itself

FIGURE 7.17 USING AN ALIAS TO JOIN A TABLE TO ITSELF

EMP_NUM	E.EMP_LNAME	EMP_MGR	M.EMP_LNAME
112	Johnson	100	Kolmycz
103	Jones	100	Kolmycz
102	Vandam	100	Kolmycz
101	Lewis	100	Kolmycz
115	Saranda	105	Williams
113	Smythe	105	Williams
111	Washington	105	Williams
107	Diante	105	Williams
106	Smith	105	Williams
104	Lange	105	Williams
116	Smith	108	Wiesenbach
114	Brandon	108	Wiesenbach
110	Genkazi	108	Wiesenbach
109	Smith	108	Wiesenbach

```
SELECT E.EMP_NUM, E.EMP_LNAME, E.EMP_MGR, M.EMP_LNAME
FROM EMP E, EMP M
WHERE E.EMP_MGR=M.EMP_NUM
ORDER BY E.EMP_MGR;
```

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ORDER BY Clause Options (1 of3)

- ORDER BY clause is especially useful when the listing order is important
  - Syntax:

    SELECT *columnlist*
    FROM *tablelist*
    [ORDER BY *columnlist* [ASC|DESC]];
  - Cascading order sequence
    - 1. ORDER BY last name
    - 2. Within matching last names, ORDER BY first name
    - 3. Within matching first and last names, ORDER BY middle initial

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ORDER BY Clause Options (2 of 3)

FIGURE 7.18 PRODUCTS SORTED BY PRICE IN ASCENDING ORDER

P_CODE	P_DESCRIPT	P_QOH	P_PRICE
54778-2T	Rat-tail file, 1/8-in. fine	43	4.99
PVC23DRT	PVC pipe, 3.5-in., 8-ft	188	5.87
SM-18277	1.25-in. metal screw, 25	172	6.99
SW-23116	2.5-in. wd. screw, 50	237	8.45
23109-HB	Claw hammer	23	9.95
23114-AA	Sledge hammer, 12 lb.	8	14.40
13-Q2/P2	7.25-in. pwr. saw blade	32	14.99
14-Q1/L3	9.00-in. pwr. saw blade	18	17.49
2238/QPD	B&D cordless drill, 1/2-in.	12	38.95
1546-QQ2	Hrd. cloth, 1/4-in., 2x50	15	39.95
1558-QVV1	Hrd. cloth, 1/2-in., 3x50	23	43.99
2232/QWE	B&D jigsaw, 8-in. blade	6	99.87
2232/QTY	B&D jigsaw, 12-in. blade	8	109.92
11QER/31	Power painter, 15 psi., 3-nozzle	8	109.99
WR3/TT3	Steel matting, 4'x8'x1/6", .5" mesh	18	119.95
89-WRE-Q	Hicut chain saw, 16 in.	11	256.99

```
SELECT P_CODE, P_DESCRIPT, P_QOH, P_PRICE
FROM PRODUCT
ORDER BY P_PRICE;
```

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ORDER BY Clause Options (3 of 3)

```
SELECT EMP_LNAME, EMP_FNAME, EMP_INITIAL, EMP_AREACODE, EMP_PHONE
FROM EMP
ORDER BY EMP_LNAME, EMP_FNAME, EMP_INITIAL;
```

	EMP_LNAME	EMP_FNAME	EMP_INITIAL	EMP_AREACODE	EMP_PHONE
1	Brandon	Marie	G	901	882-0845
2	Diante	Jorge	D	615	890-4567
3	Genkazi	Leighla	W	901	569-0093
4	Johnson	Edward	E	615	898-4387
5	Jones	Anne	M	615	898-3456
6	Kolmycz	George	D	615	324-5456
7	Lange	John	P	901	504-4430
8	Lewis	Rhonda	G	615	324-4472
9	Saranda	Hermine	R	615	324-5505
10	Smith	George	A	615	890-2984

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WHERE Clause Options (1 of 5)

- Selecting rows with conditional restrictions
  - WHERE clause is used to add conditional restrictions to the SELECT statement that limit the rows returned by the query
  - Syntax:

```
SELECT      columnlist
FROM        tablelist
[WHERE      conditionlist ]
[ORDER BY   columnlist [ASC | DESC] ];
```

- Using comparison operators on character attributes
  - May be used to place restrictions on character-based attributes
- Using comparison operators on dates
  - Date procedures are often more software-specific than other SQL procedures

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WHERE Clause Options (2 of 5)

Table 7.6 Comparison Operators	
Symbol	**Meaning**
=	Equal to
<	Less than
<=	Less than or equal to
>	Greater than
>=	Greater than or equal to
<> or !=	Not equal to

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WHERE Clause Options (3 of 5)

- Selecting rows with conditional restrictions
  - WHERE clause is used to add conditional restrictions to the SELECT statement that limit the rows returned by the query
  - Syntax:

```
SELECT      columnlist
FROM        tablelist
[WHERE      conditionlist ]
[ORDER BY   columnlist [ASC | DESC] ];
```

- Using comparison operators on character attributes
  - May be used to place restrictions on character-based attributes
- Using comparison operators on dates
  - Date procedures are often more software-specific than other SQL procedures

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WHERE Clause Options (4 of 5)

- Logical operators: AND, OR, and NOT
  - SQL allows you to include multiple conditions in a query through the use of these logical operators
  - Boolean algebra is dedicated to the use of logical operators

- Special operators
  - BETWEEN
  - IN
  - LIKE
  - IS NULL
  - NOT

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Examples (5 of 5)

```
SELECT *
FROM   PRODUCT
WHERE  P_PRICE BETWEEN 50.00 AND 100.00

SELECT *
FROM   PRODUCT
WHERE  P_PRICE > 50.00 AND P_PRICE < 100.00

SELECT P_CODE, P_DESCRIPT, V_COCE
FROM   PRODUCT
WHERE  V_CODE IS NULL

SELECT P_CODE, P_DESCRIPT, P_INDATE
FROM   PRODUCT
WHERE  P_INDATE IS NULL

SELECT *
FROM   PRODUCT
WHERE  V_CODE IN (21344, 24288)
```

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Aggregate Processing (1 of 4)

- Takes a collection of rows and reduces it to a single row
  - SQL provides useful aggregate functions that count, find minimum and maximum values, calculate averages, etc.
- Aggregate functions
  - Count
  - MIN and MAX
  - SUM and AVG
- Grouping data
  - GROUP BY clause syntax:

```
SELECT      columnlist
FROM        tablelist
[WHERE      conditionlist ]
[GROUP BY   columnlist ]
[ORDER BY   columnlist [ASC | DESC] ];
```

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Aggregate Processing (2 of 4)

Table 7.7 Some Basic SQL Aggregate Functions	
Function	**Output**
COUNT	The number of rows containing non-null values
MIN	The minimum attribute value encountered in a given column
MAX	The maximum attribute value encountered in a given column
SUM	The sum of all values for a given column
AVG	The arithmetic mean (average) for a specified column

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Aggregate Processing (3 of 4)

- HAVING clause
  - Operates very much like the WHERE clause in the SELECT statement
  - HAVING clause is applied to the output of a GROUP BY operation
  - Syntax:

```
SELECT      columnlist
FROM        tablelist
[WHERE      conditionlist ]
[GROUP BY   columnlist ]
[HAVING     conditionlist ]
[ORDER BY   columnlist [ASC | DESC] ];
```

```
SELECT V_CODE, SUM(P_QOH * P_PRICE) AS TOTCOST
FROM PRODUCT
GROUP BY V_CODE
HAVING (SUM(P_QOH * P_PRICE) > 500)
ORDER BY SUM(P_QOH * P_PRICE) DESC;
```

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Aggregate Processing (4 of 4)

- SELECT COUNT(DISTINCT V_CODE) FROM PRODUCT;
- SELECT MAX(P_PRICE) FROM PRODUCT;
- SELECT MIN(P_PRICE) FROM PRODUCT;
- SELECT AVG(P_PRICE) FROM PRODUCT;
- SELECT SUM(CUS_BALANCE) AS TOTBALANCE FROM CUSTOMER;

	TOTBALANCE
1	2619.28

- SELECT SUM(P_QOH*P_PRICE) AS TOTVALUE FROM PRODUCT;

	TOTVALUE
1	15422.02

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Subqueries (1 of 3)

- Key characteristics
  - A subquery is a query (SELECT statement) inside another query
  - A subquery is normally expressed inside parentheses
  - The first query in the SQL statement is known as the outer query
  - The query inside the SQL statement is known as the inner query
  - The inner query is executed first
  - The output of an inner query is used as the input for the outer query
  - The entire SQL statement is sometimes referred to as a nested query
- Subquery can return one or more values
  - One single value (one column and one row)
  - A list of values (one column and multiple rows)
  - A virtual table (multicolumn, multirow set of values)

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Subqueries (2 of 3)

- WHERE subqueries
  - Most common type of subquery uses an inner SELECT subquery on the right side of a WHERE comparison expression
- IN subqueries
  - IN operator: used to compare a single attribute to a list of values
  - IN subquery: values are not known beforehand, but can be derived using a query
- HAVING subqueries
  - HAVING clause: used to restrict the output of a GROUP BY query by applying conditional criteria to the grouped rows
- Multirow subquery operators: ALL and ANY
  - ALL operator compares a single value with a list of values returned by the first subquery using a comparison operator other than equals
  - ANY operator compares a single value to a list of values and select only the rows greater than or less than any value in the list

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Subqueries (3 of 3)

- FROM subqueries
  - FROM clause specifies the table(s) from which the data will be drawn
- Attribute list subqueries
  - Inline subquery: subquery expression
    - Example: can be used to list the difference between each product's price and the average product price
- Correlated subquery
  - Executes once for each row in the outer query
  - Inner query is related to the outer query; the inner query references a column of the outer subquery
  - Can also be used with the EXISTS special operator
    - Can be used whenever there is a requirement to execute a command based on the result of another query
    - Can be used with uncorrelated subqueries, but it is almost always used with correlated subqueries

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Subqueries

```
SELECT V_CODE, V_NAME FROM VENDOR
WHERE V_CODE IN (SELECT V_CODE FROM PRODUCT);
```

```
   V_CODE V_NAME
--------- ----------------------------
    21225 Bryson, Inc.
    21231 DeeEee Supply
    21344 Gomez Bros.
    23119 Randsets Ltd.
    24288 ORDVA, Inc.
    25595 Rubicon Systems

6 rows selected.
```

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Subqueries

```
SELECT CUS_CODE, CUS_LNAME, CUS_FNAME
FROM CUSTOMER
WHERE EXISTS (SELECT CUS_CODE FROM INVOICE
                    WHERE INVOICE.CUS_CODE = CUSTOMER.CUS_CODE);
```

CUS_CODE	CUS_LNAME	CUS_FNAME
10014	Orlando	Myron
10011	Dunne	Leona
10012	Smith	Kathy
10018	Farriss	Anne
10015	O'Brian	Amy

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SQL Functions

- SQL functions are very useful tools
  - Many types
- Date and time functions
  - All date functions take one parameter of a date or character data type and return a value; refer to Table 7.10
- Numeric functions
  - Can be grouped in many different ways, such as algebraic, trigonometric, and logarithmic; refer to Table 7.11
- String functions
  - Among the most-used functions in programming; refer to Table 7.12
- Conversion functions
  - Allow you to take a value of a given data type and convert it to the equivalent value in another data type; refer to Table 7.13

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Relational Set Operators (1 of 2)

- UNION
  - Combines rows from two or more queries without including duplicate rows
  - Syntax:

    query UNION *query*
- UNION ALL
  - Used to produce a relation that retains the duplicate rows
  - Used to unite more than just two queries
- INTERSECT
  - Can be used to combine rows from two queries, returning only the rows that appear in both sets
  - Syntax:

    query INTERSECT *query*

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Relational Set Operators (2 of 2)

- EXCEPT (MINUS)
  - Combines rows from two queries and returns only the rows that appear in the first set but not in the second
  - Syntax:

    query EXCEPT *query*

    and

    query MINUS *query*
- Syntax alternatives
  - Alternative syntax used to achieve the same output

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Crafting SELECT Queries

- Know your data
  - The importance of understanding the data model that you are working in cannot be overstated
  - Real-world databases are messy; most database systems remain in service in an organization for decades
- Know the problem
  - Understand the question you are attempting to answer
  - Information reporting requests will come from a range of sources; may be one-time events or ongoing operations within an application
- Build one clause at a time
  - FROM
  - WHERE
  - GROUP BY
  - HAVING
  - SELECT
  - ORDER BY

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KEY TERMS

alias	FROM	recursive query
AND	GROUP BY	rules of precedence
AVG	HAVING	SELECT
BETWEEN	IN	set-oriented
Boolean algebra	IS NULL	subquery
cascading order sequence	LIKE	SUM
correlated subquery	MAX	transaction
COUNT	MIN	WHERE
cross join	NOT	wildcard character
DISTINCT	OR	
EXISTS	ORDER BY	

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Summary (1 of 2)

- SQL commands can be divided into two overall categories: data definition language (DDL) commands and data manipulation language (DML) commands
- The ANSI standard data types are supported by all RDBMS vendors in different ways
  - The basic data types are NUMBER, NUMERIC, INTEGER, CHAR, VARCHAR, and DATE
- The SELECT statement is the main data retrieval command in SQL
- The column list represents one or more column names separated by commas
- Operations that join tables can be classified as inner joins and outer joins
- A natural join returns all rows with matching values in the matching columns and eliminates duplicate columns
- Joins may use keywords such as USING and ON
- The ORDER BY clause is used to sort the output of a SELECT statement

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Summary (2 of 2)

- The WHERE clause can be used with the SELECT, UPDATE, and DELETE statements to restrict the rows affected by the DDL command
- Aggregate functions (COUNT, MIN, MAX, and AVG) are special functions that perform arithmetic computations over a set of rows
- Subqueries and correlated queries are used when it is necessary to process data based on other processed data
- Most subqueries are executed in a serial fashion
- SQL functions are used to extract or transform data
- SQL provides relational set operators to combine the output of two queries to generate a new relation
- Crafting effective and efficient SQL queries requires a great deal of skill

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